

Deepfake Detection Challenge Teacher Notes

TOK Cognitive Science Activity Bank • Knowledge and Technology • 25-35 min

Category	Knowledge and Technology	Time	25-35 min
Difficulty	Medium	ID	deepfake-detection-challenge

Big idea: Evidence standards change when media is synthetic.

Overview

Students consider how synthetic media changes evidence standards.

Student Prompt

Inspect Deepfake Detection Challenge Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Deepfake Detection Challenge Stimulus A	Student-facing stimulus 1: Three image descriptions: authentic-looking, edited-looking, and ambiguous. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Deepfake Detection Challenge Stimulus B	Student-facing stimulus 2: A checklist for source, context, lighting, body geometry, and independent corroboration. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Use this for comparison, a second round, or a partner challenge.
Deepfake Detection Challenge Reveal / Twist	Student-facing stimulus 3: A debrief comparing technical clues with social trust cues. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Teacher reveal: connect the changed response to the big idea — Evidence standards change when media is synthetic.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Deepfake Detection Challenge Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Deepfake Detection Challenge Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	What happens to knowledge when evidence can be easily fabricated?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how evidence standards change when media is synthetic.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

- Prepare a starter stimulus using: Three image descriptions: authentic-looking, edited-looking, and ambiguous.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

Ready-to-Use Examples

- Three image descriptions: authentic-looking, edited-looking, and ambiguous.
- A checklist for source, context, lighting, body geometry, and independent corroboration.
- A debrief comparing technical clues with social trust cues.

Suggested Timing

Stage	Teacher move	Watch for
1	Show several media examples without revealing which are manipulated.	Assumptions, confidence shifts, evidence claims, and language choices.
2	Students create a checklist: source, metadata, shadows, hands, audio mismatch, reverse image search, corroboration.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Reveal answers or uncertainty.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Students discuss how much trust should change in a synthetic media environment.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Common Misconceptions

- Students may treat a tool's output as neutral because it feels automatic.
- Students may focus only on user error and miss design incentives or hidden rules.

Assessment Look-Fors

- Identifies the rule, design choice, or incentive shaping knowledge.

- Distinguishes access to information from justified knowledge.

Differentiation and Adaptations

- Short lesson: use only the first example, one response grid, and one debrief question.
- Long lesson: add a second round where students revise the method and compare outcomes.
- Low-tech option: run with printed cards, sticky notes, and a board tally.

Extension Tasks

- Design a second version of the activity that tests the same knowledge issue in another AOK.
- Find a real-world example where the same knowledge problem matters outside the classroom.
- Write a 150-word TOK exhibition-style commentary using the activity as the object context.

Related Activities

- Algorithmic Feed Simulation
- Search Engine Ranking Game
- AI Hallucination Courtroom